

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```
===== RESTART: C:/Users/ANLNA/OneDrive/Python/Python314/LAB 2 ML/LAB EXPERIMENT - 1.py (3.14.2) =====  
  
Training data:  
=====  
Sky      | AirTemp | Humidity | Wind    | Water   | Forecast | Target  
Sunny    | Warm    | Normal   | Strong  | Warm    | Same      | Yes  
Sunny    | Warm    | High     | Strong  | Warm    | Same      | Yes  
Rainy    | Cold    | High     | Strong  | Warm    | Change    | No  
Sunny    | Warm    | High     | Strong  | Cool    | Change    | Yes  
=====  
FIND-S TRACE  
=====  
Initial hypothesis h0 = ['0', '0', '0', '0', '0', '0']  
  
Example 1: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same'] -> POSITIVE  
h1 = ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']  
  
Example 2: ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same'] -> POSITIVE  
h2 = ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']  
  
Example 3: ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change'] -> NEGATIVE (ignored)  
h3 = ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']  
  
Example 4: ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change'] -> POSITIVE  
h4 = ['Sunny', 'Warm', '?', 'Strong', '?', '?']  
  
=====  
MOST SPECIFIC HYPOTHESIS (FIND-S output)  
=====  
Sky      : Sunny  
AirTemp  : Warm  
Humidity  : ?  
Wind      : Strong  
Water     : ?  
Forecast  : ?  
  
h = ['Sunny', 'Warm', '?', 'Strong', '?', '?']  
  
Classification demo:  
['Sunny', 'Warm', 'Normal', 'Strong', 'Cool', 'Change'] -> Yes  
['Rainy', 'Cold', 'Normal', 'Light', 'Warm', 'Same'] -> No
```

```
LAB EXPERIMENT - 1.py - C:/Users/ANLNA/OneDrive/ML/LAB EXPERIMENT - 1.py (3.14.2)  
File Edit Format Run Options Window Help  
  
import csv  
import sys  
  
ATTRIBUTES = ["Sky", "AirTemp", "Humidity", "Wind", "Water", "Forecast"]  
  
DATA = [  
    ["Sunny", "Warm", "Normal", "Strong", "Warm", "Same", "Yes"],  
    ["Sunny", "Warm", "High", "Strong", "Warm", "Same", "Yes"],  
    ["Rainy", "Cold", "High", "Strong", "Warm", "Change", "No"],  
    ["Sunny", "Warm", "High", "Strong", "Cool", "Change", "Yes"],  
]  
  
def load_csv(path):  
    with open(path, newline='') as f:  
        rows = [[c.strip() for c in r] for r in csv.reader(f) if r]  
        return rows[0][1:], rows[1:]  
  
def find_s(attributes, data, verbose=True):  
    n = len(attributes)  
    h = ["0"] * n  
  
    if verbose:  
        print("=" * 64)  
        print("FIND-S TRACE")  
        print("=" * 64)  
        print(f"Initial hypothesis h0 = {h}\n")  
  
    for i, row in enumerate(data, start=1):  
        x, label = row[-1], row[-1].strip().lower()  
  
        if label != "yes":  
            if verbose:  
                print(f"Example (i): {x} -> NEGATIVE (ignored)")  
                print(f"          h(i) = {h}\n")  
            continue  
  
        if verbose:  
            print(f"Example (i): {x} -> POSITIVE")
```

Lab experiment-1

```
LAB EXPERIMENT - 1.py - C:/Users/ANLNA/OneDrive/ML/LAB EXPERIMENT - 1.py (3.14.2)  
File Edit Format Run Options Window Help  
  
import csv  
  
f = open("training_data.csv", "w")  
f.write("""Sky,AirTemp,Humidity,Wind,Water,Forecast,EnjoySport  
Sunny,Warm,Normal,Strong,Warm,Same,Yes  
Sunny,Warm,High,Strong,Warm,Same,Yes  
Rainy,Cold,High,Strong,Warm,Change,No  
Sunny,Warm,High,Strong,Cool,Change,Yes  
""")  
f.close()  
  
data = list(csv.reader(open("training_data.csv")))  
header = data[0]  
data = data[1:]  
n = len(header) - 1  
  
values = []  
for i in range(n):  
    values.append(sorted(set(row[i] for row in data)))  
  
def covers(h, x):  
    for i in range(n):  
        if h[i] != "?" and h[i] != x[i]:  
            return False  
    return True  
  
def more_general(g, s):  
    for i in range(n):  
        if g[i] != "?" and g[i] != s[i] and s[i] != "0":  
            return False  
    return True  
  
S = ["0"] * n  
G = ["?"] * n  
print("Initial S:", S)  
print("Initial G:", G)  
  
for k, row in enumerate(data, 1):  
    x = row[-1:]  
    print("\nExample", k, x, row[-1])  
  
    if row[-1] == "Yes":  
        G = [g for g in G if covers(g, x)]  
        if S[0] == "0":  
            S = x[:]  
        else:  
            for i in range(n):  
                if S[i] != x[i]:  
                    S[i] = "?"  
    else:  
        newG = []  
        for g in G:  
            if not covers(g, x):  
                newG.append(g)  
        continue  
        for i in range(n):
```

```
IDLE Shell 3.14.2  
File Edit Shell Debug Options Window Help  
  
>>> == RESTART: C:/Users/ANLNA/AppData/Local/Programs/Python/Python314/LAB 2 ML.py ==  
Initial S: ['0', '0', '0', '0', '0', '0']  
Initial G: [['?', '?', '?', '?', '?', '?']]  
  
Example 1 ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same'] Yes  
S: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']  
G: [['?', '?', '?', '?', '?', '?']]  
  
Example 2 ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same'] Yes  
S: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']  
G: [['?', '?', '?', '?', '?', '?']]  
  
Example 3 ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change'] No  
S: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']  
G: [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', 'Same']]  
  
Example 4 ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change'] Yes  
S: ['Sunny', 'Warm', '?', 'Strong', '?', '?']  
G: [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?']]  
  
Final S: ['Sunny', 'Warm', '?', 'Strong', '?', '?']  
Final G: [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?']]  
>>>
```

Lab experiment-2